

Comprehensive Mathematics Curriculum

Complete Standard Course Syllabus: Ordinary Level (O'Level) & Advanced Level (A'Level)

Institution: Ejikeme Masterclass

Target Audience: High School & Pre-University

URL: ejikeme-masterclass.github.io

PART 1: O'LEVEL MATHEMATICS CURRICULUM

1. Number, Arithmetic & Algebra

1.1 Number Types & Properties

- Prime numbers, composite numbers, factors, and multiples
- Highest Common Factor (HCF) and Least Common Multiple (LCM)
- Set notation, set operations, and Venn diagrams

1.2 Fractions, Decimals & Percentages

- Conversions between fractions, decimals, and recurring decimals
- Percentage increase/decrease and reverse percentage
- Financial math: Profit/loss, simple interest, and compound interest

1.3 Indices, Surds & Standard Form

- Laws of indices (positive, negative, and fractional exponents)
- Scientific notation / standard form ($A \times 10^n$)
- Simplifying surds and rationalizing denominators

1.4 Ratio, Proportion & Estimation

- Direct and inverse variation/proportion
- Scale factors, speed, density, and unit conversions
- Rounding, significant figures, upper/lower bounds, and error limits

2. Algebra & Coordinate Geometry

2.1 Algebraic Expressions & Equations

- Expansion, factorization (quadratic and cubic), algebraic fractions
- Changing the subject of a formula
- Linear, quadratic, and simultaneous equations
- Linear inequalities in one and two variables

2.2 Functions & Sequences

- Function notation, domain, range, inverse, and composite functions
- Linear, quadratic, and special sequences
- Finding the n -th term formula

2.3 Coordinate Geometry & Graphs

- Line equations ($y = mx + c$), midpoint, gradient, distance formula
- Parallel and perpendicular line properties
- Graphical analysis of quadratic, cubic, exponential, and reciprocal functions

3. Geometry, Trigonometry, Vectors & Statistics

3.1 Geometry & Mensuration

- Angle properties, parallel lines, interior/exterior angles of polygons
- Congruence and similarity theorems
- Circle theorems (tangents, secants, cyclic quadrilaterals)
- Perimeter, area, volume, and surface area of 2D/3D shapes

3.2 Trigonometry & Vectors

- Right-angled trigonometry (SOH CAH TOA) & Pythagoras' theorem
- Sine Rule, Cosine Rule, and Area of Triangle ($\frac{1}{2}ab \sin C$)
- 2D Column vectors, magnitude, scalar multiplication, vector proofs
- Geometric transformations: Translation, reflection, rotation, enlargement

3.3 Statistics & Probability

- Frequency tables, mean, median, mode, range, grouped data estimates
- Histograms (frequency density), cumulative frequency curves, box plots
- Probability of single/combined events, tree diagrams, Venn diagrams, conditional probability

PART 2: A'LEVEL MATHEMATICS CURRICULUM

1. Pure Mathematics Core

1.1 Proof & Advanced Algebra

- Proof by exhaustion, contradiction, and mathematical induction
- Polynomial division, factor/remainder theorems, partial fractions
- Binomial expansion for rational n ($(1+x)^n$)

1.2 Functions & Coordinate Geometry

- Modulus function, transformations ($f(x+a)$, $af(x)$, $|f(x)|$)
- Circle equations $(x-a)^2 + (y-b)^2 = r^2$
- Parametric equations and Cartesian conversion

1.3 Advanced Trigonometry & Exponentials

- Radians, secant, cosecant, cotangent functions
- Double angle, compound angle formulas, and $R \sin(x \pm \alpha)$ form
- Graphs of e^x and $\ln x$, solving exponential equations

1.4 Differentiation & Integration

- Product, quotient, chain rule, implicit and parametric differentiation
- Integration by substitution, parts, partial fractions, and differential equations
- Areas under curves and volumes of revolution

1.5 Numerical Methods & 3D Vectors

- Root finding (Newton-Raphson, sign change methods), trapezium rule
- 3D vector algebra ($\mathbf{i}, \mathbf{j}, \mathbf{k}$), scalar (dot) product, vector equations of lines

2. Applied Mathematics: Mechanics & Statistics

2.1 Mechanics

- Kinematics in 1D and 2D (SUVAT equations, calculus in kinematics)
- Forces & Newton's Laws, friction ($F \leq \mu R$), connected bodies
- Moments, static equilibrium, and center of mass
- Projectile motion under gravity

2.2 Probability & Statistics

- Sampling methods, measures of central tendency and variation
- Binomial distribution ($B(n, p)$) and Normal distribution ($N(\mu, \sigma^2)$)
- Hypothesis testing, significance levels, critical regions
- Correlation testing (PMCC, Spearman's rank)

Official Syllabus Document — Ejikeme Masterclass (<https://ejikeme-masterclass.github.io>)